

Data Types – Simple Hinglish Summary

Source: <https://javascript.info/types>

Main baat

JavaScript me har value ka ek **type** hota hai (jaise string ya number). JS me **8 basic data types** hain. Ye chapter unka general intro hai, detail aage ke chapters me aayegi.

Dynamically typed language

Variable me **kisi bhi type ki value** rakh sakte ho, aur baad me type badal bhi sakte ho. Ise **"dynamically typed"** kehte hain.

```
javascript  
  
let message = "hello";  
message = 123456;    // koi error nahi
```

1. Number

Integer aur decimal (floating point), dono ke liye ek hi type.

```
javascript  
  
let n = 123;  
n = 12.345;
```

Operations: $\otimes$, $\oslash$, $\oplus$, $\ominus$ etc.

Special numeric values

- **Infinity** → mathematical infinity, har number se bada. Zero se divide karne par milta hai.

```
javascript  
  
alert( 1 / 0 );    // Infinity
```

- **NaN** (Not a Number) → computational error. Galat math operation ka result.

```
javascript  
  
alert( "not a number" / 2 );    // NaN
```

NaN "sticky" hota hai: uske saath koi bhi math karo, result NaN hi rahega.

javascript

```
alert( NaN + 1 );    // NaN
alert( 3 * NaN );    // NaN
```

(Ek hi exception: $\text{NaN} ** 0 = 1$)

Math "safe" hai

JS me zero se divide karna jaisi galtiyon par script **crash nahi hoti**. Worst case me NaN milta hai.

2. BigInt

- Number type $\pm(2^{53} - 1)$ (yaani 9007199254740991) tak ke integers safely store karta hai. Isse bade numbers me **precision error** aata hai.

javascript

```
console.log(9007199254740991 + 1); // 9007199254740992
console.log(9007199254740991 + 2); // 9007199254740992 (same!)
```

- BigInt** bahut bade integers (kisi bhi length ke) ke liye hai, jaise cryptography ke liye.
- Integer ke end me n lagao:

javascript

```
const bigInt = 1234567890123456789012345678901234567890n;
```

- Kam use hota hai, alag chapter me detail hai.

3. String

String ko **quotes** me likhna zaruri hai. **3 tarah ke quotes**:

Quote	Example	Khaasiyat
Double $" "$	$"\text{Hello}"$	Simple quote
Single $' '$	$'\text{Hello}'$	Simple quote (double jaisa hi)
Backtick $` `$	$\`\text{Hello}\`$	Extended functionality

Backticks me variable/expression embed kar sakte ho $\${\dots}$ se:

javascript

```
let name = "John";

alert( `Hello, ${name}!` );           // Hello, John!
alert( `the result is ${1 + 2}` );    // the result is 3
```

Ye **sirf backticks me** kaam karta hai. Double quotes me nahi:

javascript

```
alert( "the result is ${1 + 2}" );    // the result is ${1 + 2}
```

Note: JS me alag se **character type nahi hai** (C/Java jaisa `char`). Sirf `string` hai, chahe 0, 1 ya zyada characters ho.

4. Boolean

Sirf **do values**: `true` aur `false`. Yes/no ke liye use hota hai.

javascript

```
let nameFieldChecked = true;
let ageFieldChecked = false;

let isGreater = 4 > 1;
alert( isGreater );    // true
```

Comparison ka result bhi boolean hota hai.

5. `null`

- Ye ek **alag type** hai jisme sirf ek value `null` hai.
- Matlab: "**kuch nahi**", "**empty**" ya "**value unknown**".
- Ye "non-existing object ka reference" nahi hai (dusri languages ki tarah).

javascript

```
let age = null;    // age unknown hai
```

6. `undefined`

- Ye bhi **alag type** hai. Matlab: "**value assign nahi hui**."
- Variable declare kiya lekin value nahi di, to value `undefined` hoti hai.

javascript

```
let age;
alert(age); // "undefined"
```

- Explicitly `undefined` assign kar sakte ho, lekin **recommend nahi**.
- **Rule:** "empty/unknown" dikhane ke liye `null` use karo. `undefined` sirf unassigned cheezon ke default ke liye rakho.

7. Object aur Symbol

- **Object:** Baaki sab types **primitive** hain (ek hi cheez store karte hain). Object **collections of data** aur complex cheezein store karta hai. Detail baad me.
- **Symbol:** Objects ke liye **unique identifiers** banane ke liye. Detail baad me.

8. `typeof` operator

Value ka type batata hai (string ke roop me).

```
javascript

typeof undefined // "undefined"
typeof 0          // "number"
typeof 10n        // "bigint"
typeof true       // "boolean"
typeof "foo"      // "string"
typeof Symbol("id") // "symbol"
typeof Math       // "object"
typeof null       // "object"    <-- language ki galti
typeof alert      // "function"
```

Dhyan dene wali baatein

- `typeof null` = `"object"`: Ye JS ki **purani officially recognized galti** hai, compatibility ke liye rakhi gayi hai. `null` object **nahi hai**.
- `typeof alert` = `"function"`: Functions asal me **object type** ke hi hain, lekin `typeof` unhe alag dikhata hai (practical hai).
- `typeof x` aur `typeof(x)` same hain. `typeof` ek **operator** hai, function nahi. Brackets sirf grouping ke hain. `typeof x` zyada common hai.

Summary

8 data types:

7 Primitive:

Type	Kis liye
number	Integer aur decimal (limit: $\pm(2^{53}-1)$)
bigint	Kisi bhi length ke integers
string	Text (alag char type nahi)
boolean	true / false
null	Unknown / empty value
undefined	Unassigned value
symbol	Unique identifiers

1 Non-primitive:

Type	Kis liye
object	Complex data structures

Practice Task

String quotes ka output kya hoga?

```
javascript
let name = "Ilya";

alert( `hello ${1}` );           // hello 1
alert( `hello ${"name"}` );      // hello name
alert( `hello ${name}` );        // hello Ilya
```

Backticks me `${...}` ke andar ka expression evaluate hokar string me judta hai.